

Mitigation of Sensitive Data Exfiltration in LLM-Based Code Interpreters

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Abstract: We introduce a dynamic flow tracking framework that restricts output channels when sensitive source code or keys are loaded into context.

1. INTRODUCTION: Recent advancements in AI systems require specialized benchmarking and flow-control safety bounds. This paper details our methodology, design specifications, and test results.
2. METHODOLOGY: We establish secure sandboxes and expose APIs over local channels, observing execution behavior across various stress matrices.
3. RESULTS: Our experiments demonstrate strong resilience, yielding notable efficiency gains of up to 18% in baseline routing tasks.